

# ECON 002: Principles of Macroeconomics

- Lecture 4: GDP and Stable Prices



GDP: the market value of all final goods and services **produced** for a marketplace during a period of time, within a country's borders

- “Produced”
- Many things are “bought” by individuals, but they are not “produced”
- Examples: Bonds, Stocks, Land



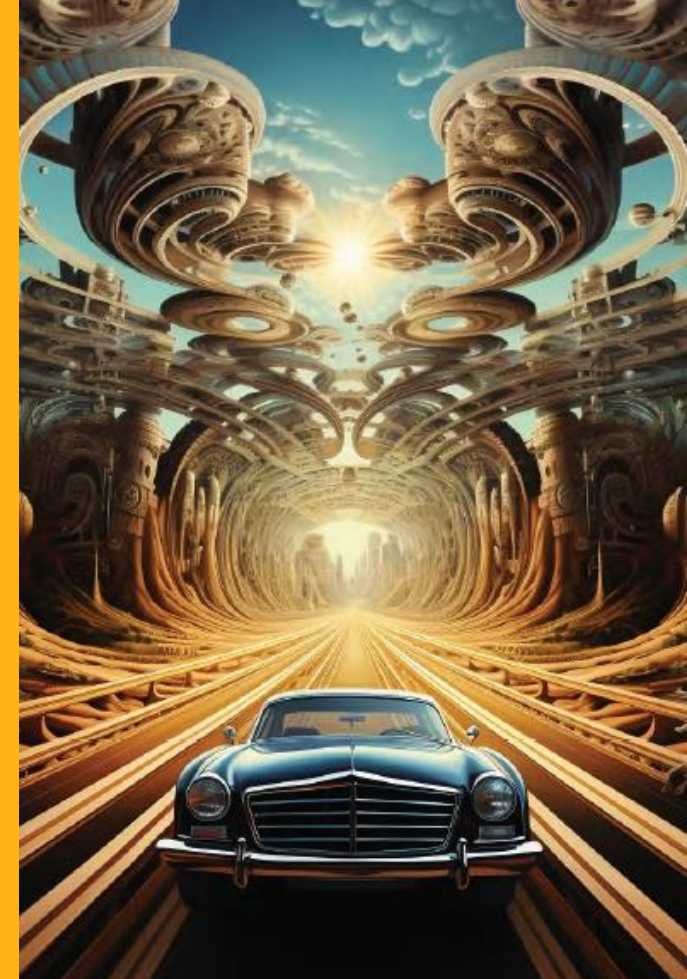
GDP: the market value of all final goods and services produced **for a marketplace** during a period of time, within a country's borders



- “For a Marketplace”
- Marketplace = defined location (e-location) where goods and services are bought and sold
- What goods and services are not bought or sold in a formal marketplace?
  - Housework
  - Yard Work
  - Child Care
  - Volunteering
- Informal marketplace
  - Illegal activities
  - “Trading services”

GDP: the market value of all final goods and services produced for a marketplace during a period of time, within a country's borders

- “During a Period of Time”
- Typically define a period of time as one year, 2024
- Everything produced in 2024 counts towards 2024 GDP
- A used 2017 Toyota Camry is sold on Craigslist in April 2024.
- Is the car counted in 2017 or 2024 GDP?
- A used 2017 Toyota Camry is sold at a used car dealership in April 2024
- What is counted in 2017 GDP? 2024 GDP?



GDP: the market value of all final goods and services produced for a marketplace during a period of time, **within a country's borders**



- “Within a Country’s Borders”
- GDP: Production within the geographical border of a country.
- Can also be a City, State, Continent or Planet!
- Gross National Product (GNP): Production by citizens of a country, no matter where they are

# Measuring GDP

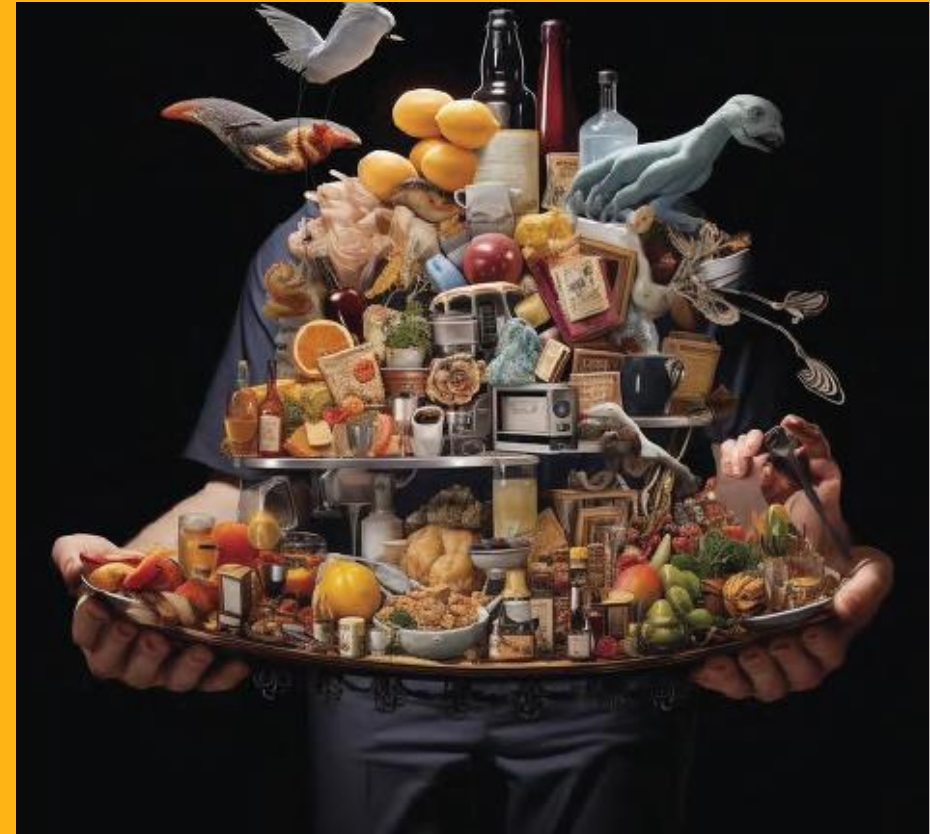


- Most common way to measure GDP is the **Expenditure Approach**
- Bureau of Economic Analysis ([www.bea.gov](http://www.bea.gov))  
National Income and Product Accounts (NIPA) table
- Four General Categories
  1. Consumptions of goods and services (C): purchases by the households
  2. Private investment of goods and services (I): purchases by firms/businesses
  3. Government goods and services (G): purchase by governments
  4. Net Exports (NX) = Exports (X) – Imports (M): net purchases by foreigners
    1. Exports (X): Goods sold by home country, money **comes from** abroad
    2. Imports (M): Goods bought by home country, money **goes** abroad
- GDP defined as  **$Y = C + I + G + NX$**

$$Y = C + I + G + NX$$

- Consumption (C)

- Consumption (67%): a final good or service purchased by a household
- Goods (21%): a tangible item that consumers gain ownership over when purchasing
  - Durable Goods (9.5%): Motor Vehicles, Furniture, Recreation Goods
  - Non-Durable Goods (14.9%): Food, Clothing, Gasoline
- Services (47%): an intangible good that we purchase but do not gain ownership of: health care, airline tickets, financial services, etc.



# Incorporating Housing

- Are homes a durable good?
- **New homes** are considered an *investment*  
Household buying a home ~ Firm purchasing a plant
- Households do not take their home when they move
- **Rental Housing**: counted as a service
- **Imputed Rent**: the cost to you if you rented your home
  - If your home would rent for \$1,000/month = \$12,000/year, count \$12,000 in “imputed rent” as a service
- Housing Services make up 12% of the total GDP!



$$Y = C + I + G + NX$$



## Investment (I)

- Investment (18%): goods and services purchased by firms
- Nonresidential Investment (13.9%): Capital machinery and plants
  - Tractor, Assembly Line Part, Computer, Software, Desk, Building/Plant
- Residential Investment (4.0%): New home construction
- Change in Private Inventories (~0%)

# Change in Private Inventories

## Campus Bookstore in 2024

\$10,000 worth of hoodies made and up for sale!

GDP in 2024 increases by \$10,000

Sell \$8,000 worth of hoodies

Consumption in 2024 increases by???

\$2,000 worth of hoodies unsold at the end of the year

Added to inventory/stock room/etc.

Increase inventory in 2024 = increase investment in 2024



## Campus Bookstore in 2025

Remaining \$2,000 in hoodies sold!

Consumption in 2025 increases by \$2,000

Where did the hoodies come from?

Inventory/stock room/etc. (not produced)

Decrease in inventory of \$2,000

Change in Inventory = -\$2,000

Investment decreases by \$2,000 in 2025

Total Change in 2025 GDP? \$0

Consumption = +\$2,000, Investment = -\$2,000

$$Y = C + I + \mathbf{G} + NX$$

## Government Purchases (G)

- Government Purchases (17.2%): Goods and services that the government buys
- Federal Spending (6.5%): bought by the Federal Government
  - National Defense (3.7%)
  - What else?
- State and Local Spending (10.7%): bought by local governments
- Does not include transfer payments such as Social Security, Unemployment Benefits, Welfare. Why?



$$Y = C + I + G + NX$$



## Net Exports (NX)

- Net Exports (NX) = Exports (X) minus Imports (M) =  $X - M$
- Exports (X) : Goods and services purchased BY foreign entities
- Imports (M) : Goods and services bought FROM foreign entities
- Exports are 10.8% of GDP
- Imports are 13.2% of GDP
- Net Exports are  $10.8\% - 13.2\% = -2.4\%$  of GDP

$$Y = C + I + G + NX$$



US Nominal GDP in 2025 Q3

$C = 21.11$  Trillion

$I = 5.42$  Trillion

$G = 5.32$  Trillion

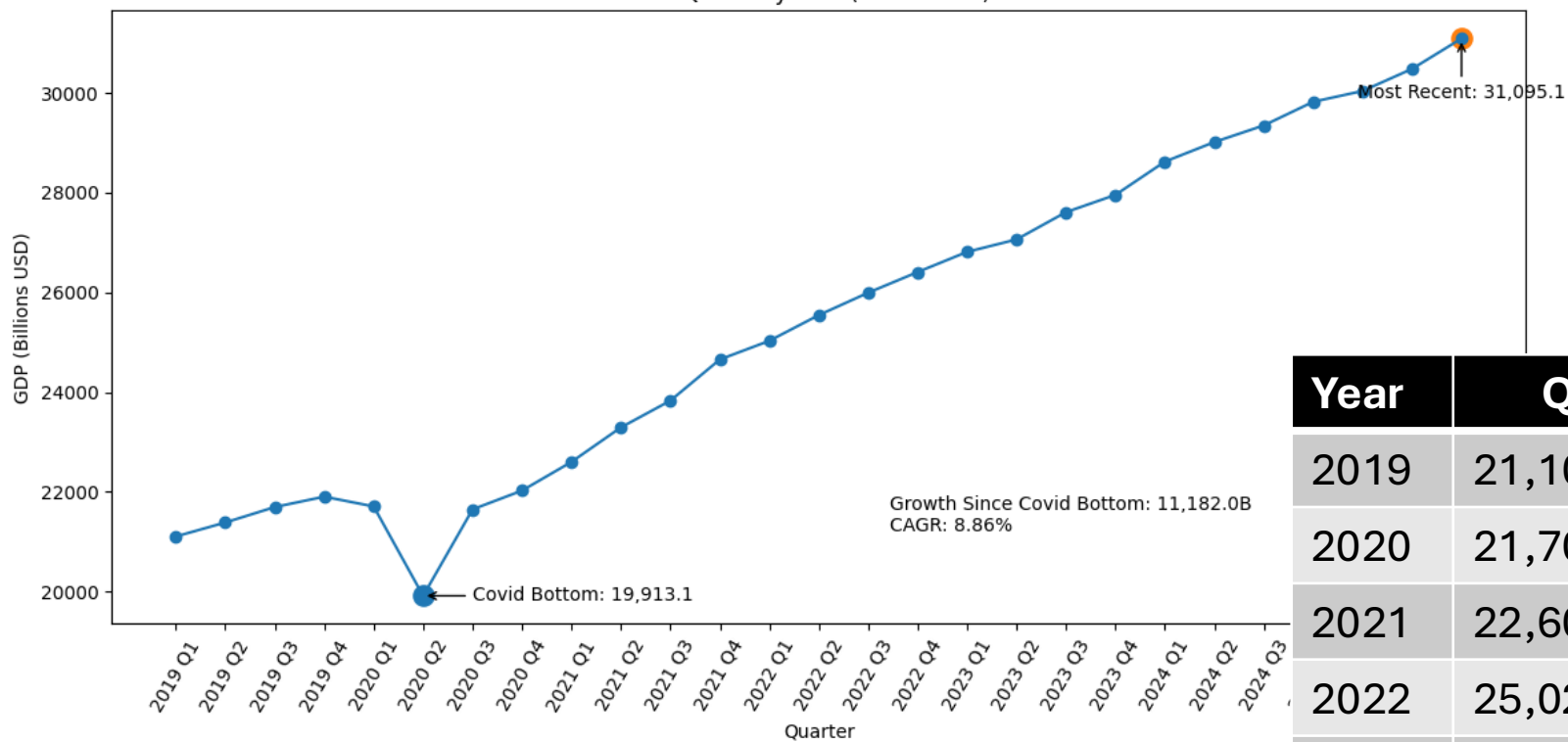
$X = 3.36$  Trillion

$M = 4.12$  Trillion

$$Y = 21.11 + 5.42 + 5.32 + (3.36 - 4.12) = 31.095 \text{ Trillion}$$

# Total US GDP

US Quarterly GDP (2019-2025)



| Year | Q1       | Q2       | Q3       | Q4       |
|------|----------|----------|----------|----------|
| 2019 | 21,104.1 | 21,384.8 | 21,694.3 | 21,902.4 |
| 2020 | 21,706.5 | 19,913.1 | 21,647.6 | 22,024.5 |
| 2021 | 22,600.2 | 23,292.4 | 23,829.0 | 24,654.6 |
| 2022 | 25,029.1 | 25,544.3 | 25,994.6 | 26,408.4 |
| 2023 | 26,813.6 | 27,063.0 | 27,610.1 | 27,957.0 |
| 2024 | 28,624.1 | 29,016.7 | 29,349.9 | 29,825.2 |
| 2025 | 30,042.1 | 30,485.7 | 31,095.1 |          |

# State GDP by Country

## Comparing U.S. States to Entire Countries by GDP



### Article & Sources:

<https://howmuch.net/articles/comparing-us-states-to-entire-countries-by-gdp-2019>

GDP by U.S. State - Bureau of Economic Analysis - <https://www.bea.gov>

GDP by Country - International Monetary Fund - <https://www.imf.org>

howmuch.net

# Alternative Measures of GDP



- Remember Best Buy:
  - Step 1: Raw materials are gathered, sold to parts manufacturer for \$50
  - Step 2: Parts are created by manufacturer and sold to Dell for \$150
  - Step 3: Dell assembles the computer and sells to Best Buy for \$350
  - Step 4: Best Buy sells computer to you for \$400
- We only add the final \$400 sale to GDP (do not double count!)
- Each stage of the production process adds value to the final product
  - Step 1: \$50 in raw materials sold to parts manufacturer (\$50 in value)
  - Step 2: \$50 in raw materials sold becomes \$150 worth of parts (\$100 in value-added)
  - Step 3: \$150 in parts assembled and sold for \$350 (\$200 in value-added)
  - Step 4: \$350 computer sold for \$400 at central location (\$50 in added-value)
- Each step adds value to the product = “Profit” of the firm
- Value-Added GDP =  $50 + 100 + 200 + 50 = \textbf{\$400}$
- Value-Added GDP = Firm Profit

# Alternative Measures of GDP

- Best Buy made \$50. Who gets that \$50?
  - Labor = Income
  - Rent = Landlord Income
  - Interest Payment = Bank/Lender = Income
  - Owner = Income
- Factor Payments GDP = Household Income!
- $GDP = Y =$ 
  - Total Production (Expenditure Approach)
  - Firm Profit (Value-Added Approach)
  - Household Income (Factor Payments Approach)

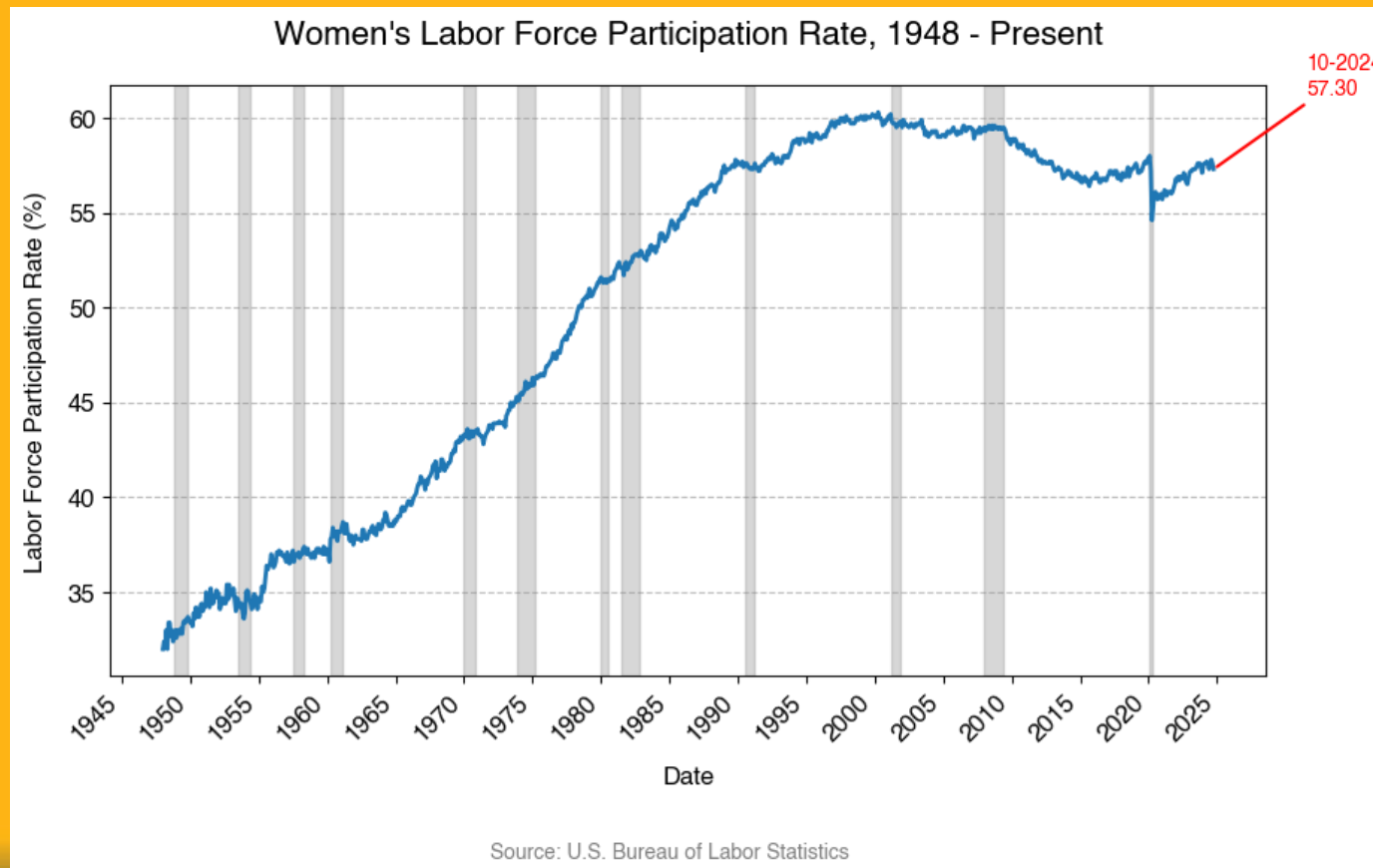


# Shortcomings of GDP

**Household Activities:** GDP ignores vital household activities like childcare and home maintenance

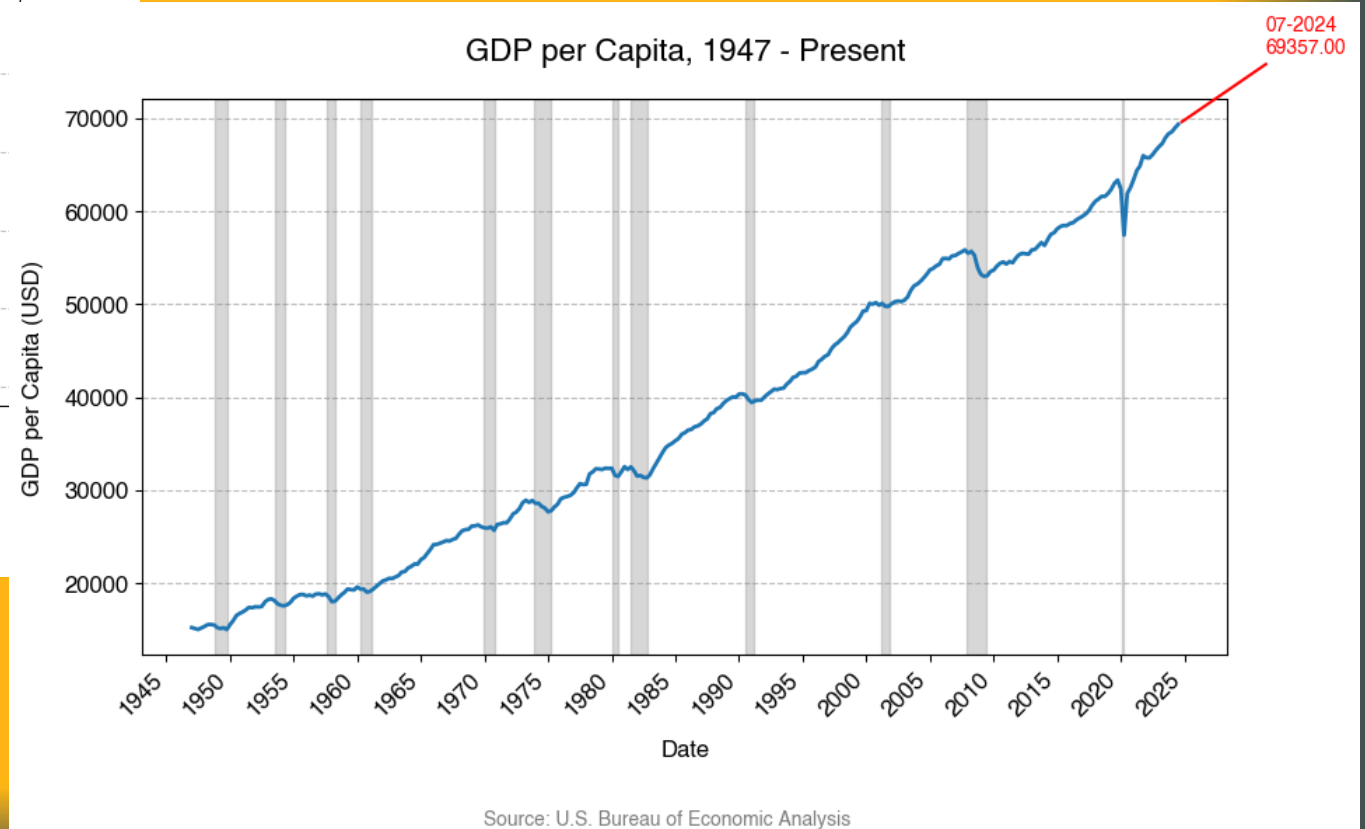
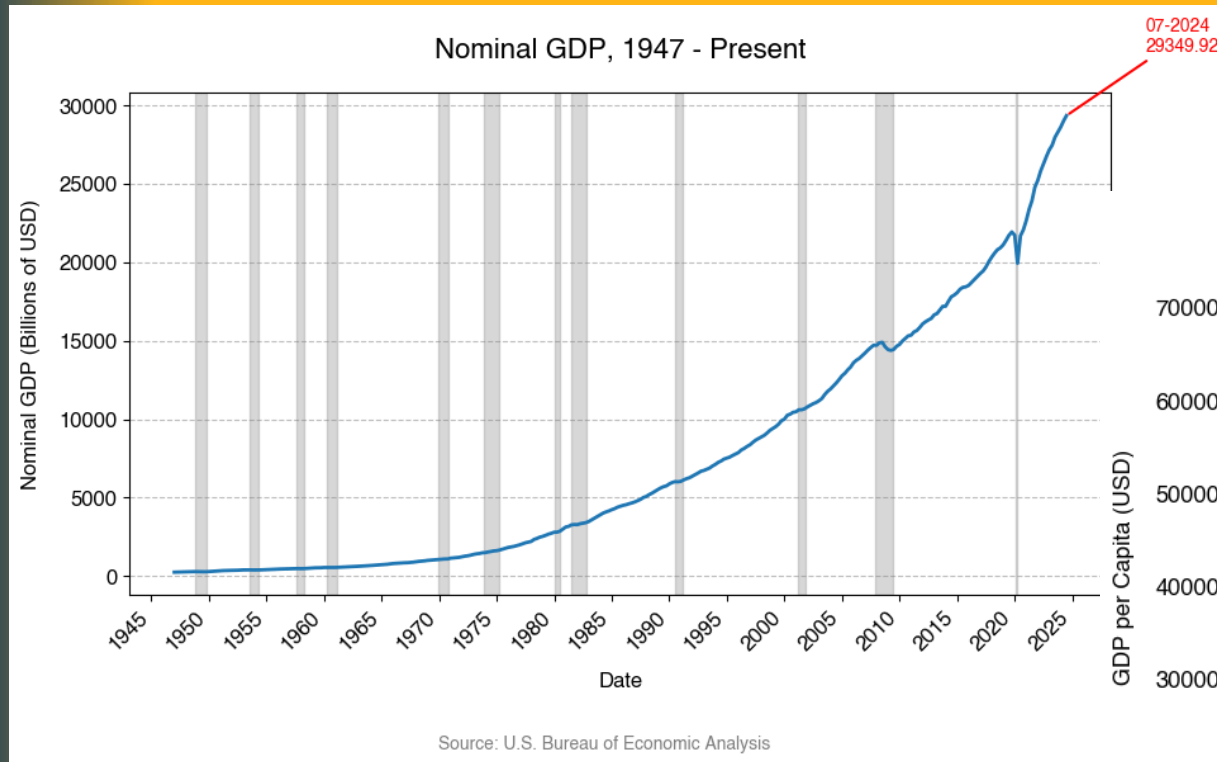
**Illegal Sales and Work:** GDP does not capture illegal sales and work in the black market or underground economy.

**Changing Labor Force Dynamics:** GDP overlooks gig economy and changes in the characteristics of workers and jobs



# GDP and Well Being?

GDP/Capita, Inequality, Health, Gross National Happiness (GNH)



# Calculating GDP

- Simple UCSB Economy

|         | 2009     |       |
|---------|----------|-------|
| Product | Quantity | Price |
| Food    |          |       |
| Housing |          |       |
| Movies  |          |       |

# Calculating GDP

- Simple UCSB Economy

|         | 2009     |       | 2016     |       |
|---------|----------|-------|----------|-------|
| Product | Quantity | Price | Quantity | Price |
| Food    |          |       |          |       |
| Housing |          |       |          |       |
| Movies  |          |       |          |       |

# Calculating GDP

- Simple UCSB Economy

|         | 2009     |       | 2016     |       |
|---------|----------|-------|----------|-------|
| Product | Quantity | Price | Quantity | Price |
| Food    | 1000     | 1     | 1500     | 2     |
| Housing | 100      | 100   | 150      | 150   |
| Movies  | 500      | 5     | 1000     | 10    |

# Calculating GDP

- Nominal GDP: Value of final goods and services evaluated at ***current year prices***
  - Total dollars exchanged
- Real GDP: Value of final goods and services evaluated at ***base year prices***
  - Value of dollars exchanged if prices did not change from the prices in an arbitrary (base) year
- GDP Deflator: Measure of how much nominal GDP changes are driven by price changes

# Using the Deflator to Calculate Price Increases

- How much did prices increase between 2009 and 2017?
- How much did real GDP increase between 2009 and 2017?

# Real GDP vs Nominal GDP with a single good

| Movie                                  | World Gross<br>(Nominal) | Movie |  |
|----------------------------------------|--------------------------|-------|--|
| 1. Avatar (2009)                       | \$2,923,706,026          |       |  |
| 2. Avengers: Endgame (2019)            | \$2,799,439,100          |       |  |
| 3. Avatar: The Way of Water (2022)     | \$2,320,250,281          |       |  |
| 4. Titanic (1997)                      | \$2,264,743,305          |       |  |
| 5. Star Wars: The Force Awakens (2015) | \$2,071,310,218          |       |  |
| 6. Avengers: Infinity War (2018)       | \$2,052,415,039          |       |  |
| 7. Spider-Man: No Way Home (2021)      | \$1,921,847,111          |       |  |
| 8. Jurassic World (2015)               | \$1,671,537,444          |       |  |
| 9. The Lion King (2019)                | \$1,663,075,401          |       |  |
| 10. The Avengers (2012)                | \$1,520,538,536          |       |  |

# Real GDP vs Nominal GDP with a single good

| Movie                                  | Worldwide Gross (Nominal) |  | Movie                                  | Ticket Sales |
|----------------------------------------|---------------------------|--|----------------------------------------|--------------|
| 1. Avatar (2009)                       | \$2,923,706,026           |  | 1. Gone w/the Wind (1939)              | 202,044,600  |
| 2. Avengers: Endgame (2019)            | \$2,799,439,100           |  | 2. Star Wars (1977)                    | 178,119,600  |
| 3. Avatar: The Way of Water (2022)     | \$2,320,250,281           |  | 3. Sound of Music (1965)               | 142,415,400  |
| 4. Titanic (1997)                      | \$2,264,743,305           |  | 4. ET (1982)                           | 141,854,300  |
| 5. Star Wars: The Force Awakens (2015) | \$2,071,310,218           |  | 5. Titanic (1997)                      | 135,474,500  |
| 6. Avengers: Infinity War (2018)       | \$2,052,415,039           |  | 6. The Ten Commandments (1956)         | 131,000,000  |
| 7. Spider-Man: No Way Home (2021)      | \$1,921,847,111           |  | 7. Jaws (1975)                         | 128,078,800  |
| 8. Jurassic World (2015)               | \$1,671,537,444           |  | 8. Doctor Zhivago (1965)               | 124,135,500  |
| 9. The Lion King (2019)                | \$1,663,075,401           |  | 9. Exorcist (1973)                     | 110,599,200  |
| 10. The Avengers (2012)                | \$1,520,538,536           |  | 10. Snow White and the 7 Dwarfs (1937) | 109,000,000  |

# Stable Prices

## Goal 2: Stable Prices

Inflation = Prices are rising

Deflation = Prices are falling

How much do consumers care about all prices?

What is the ideal level of inflation/deflation?



# Federal Reserve's "Stable Prices"

**Why is the inflation target not higher?**

Today \_\_\_\_\_ 6 Months \_\_\_\_\_ 1 Year

Invest, Prepare \_\_\_\_\_ Produce \_\_\_\_\_ Sell at Expected Price

As prices rise, workers expect cost-of-living adjustments/wage increases

Example: Inflation is 25% away than expectations.

If inflation expectations are 2%, inflation range is 1.5% to 2.5%

If inflation expectations are 50%, inflation ranges from 37.5% to 62.5%

At 37.5% inflation, firms don't sell at high enough prices to pay input costs.

At 62.5% inflation, workers don't get raises to match the increasing prices of goods.



# Federal Reserve's "Stable Prices"

## Why is the inflation target not lower?

### Deflation:

1. Consumer Response: When the prices of goods and services are expected to decrease, consumers delay purchases.

Less spending = Less production = Lower GDP = Unemployment

### 2. Debt Burden:

Situation: \$50,000 loan today, have to payback in 25 years  
\$50,000 in 25 years is equivalent to \$30,477 today at 2% inflation.  
Wages, prices, etc. rise, so the debt burden (\$50,000) is less costly over time.

If prices are falling and interest rates don't adjust, debt burden is growing! Deflation-still owe \$50,000, but wage, prices, etc. have decreased! At 2% deflation for 25 years, \$50,000 today is equivalent to \$82,000 in 25 years!

### 3. Wage Adjustments

$\text{Real Wage Growth} = \text{Nominal Wage Growth} - \text{Inflation Rate}$

Easier to decrease real wage than nominal wage



# Consumer Price Index

GDP Deflator Growth: measure of inflation

Do the prices of ALL goods and services matter to consumers?



# Consumer Price Index

- How do we calculate an “average price” for all goods purchased by consumers?
- **STEP 1: What do consumers purchase?**
- Determine basket of goods:



# Consumer Price Index

- How do we calculate an “average price” for all goods purchased by consumers?
- **STEP 2: How heavy is each good in the basket?**
- Determine the weight of goods and services in the basket:



# Consumer Price Index

- How do we calculate an “average price” for all goods purchased by consumers?
- **STEP 3: Calculate the weighted cost of a basket**
- Similar to GDP, calculate the cost of the average good bought by a household



# Consumer Price Index

**Calculate Base Year (BY) Basket Cost:**  
**BY Prices x BY Q**

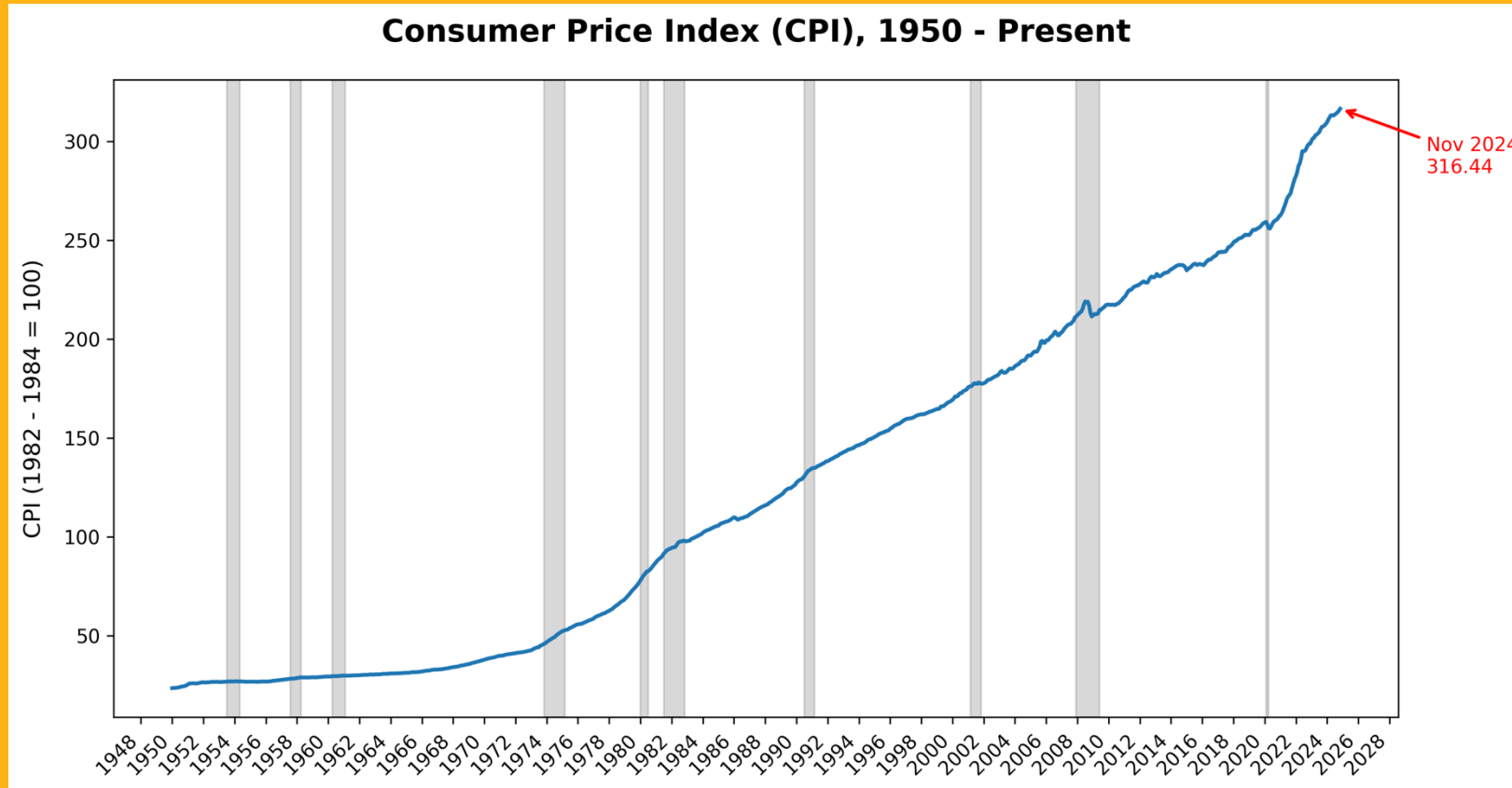
**Calculate Current Year (CY) Basket Cost: CY Prices x BY Q**

**CPI = Ratio of Basket Costs**

**Inflation = CPI Growth Rate**



# Consumer Price Index



**Base Year: 1982-1984 Average Weights/Prices, CPI = 100**

**Today (November, 2024), CPI = 316.44**

**1950 CPI = 24**

# Is the CPI an accurate measure of inflation?

## Some Potential Problems with CPI

**Substitution, quality, and new products can distort inflation measurements.**

**Consumers may change buying habits and price hikes may not always correlate with improved quality.**

**CPI doesn't cover new products or cheap stores/websites.**

**Economists believe it overestimates inflation by 0.5-1%.**

